

ASSESSMENT TOOL 3 – Resolution Algorithm Implementation

NAME : RAGAVI S

REGISTER NO : 192565027

COURSE CODE : CSA1717

COURSE NAME : ARTIFICIAL INTELLIGENCE

Question 1 — Rain and Wet Ground

Symbols:

- R = It rains
- W = Ground is wet

1. Propositional Logic Representation

- $R \rightarrow W$
- R

2. CNF Conversion

- $R \rightarrow W \Rightarrow \neg R \vee W$ (implication eliminated; already in CNF)
- R (unit clause, already CNF)

Negate the Goal

Goal: $W \rightarrow$ Negated Goal: $\neg W$

Resolution Steps

#	Clause	Source
1	$\neg R \vee W$	Premise (CNF)
2	R	Fact
3	$\neg W$	Negated Goal

Step 1: Resolve (1) & (2): $\neg R \vee W + R \rightarrow W$

Step 2: Resolve W & $\neg W$ (3) $\rightarrow$ EMPTY CLAUSE ($\square$)

Conclusion: Contradiction derived $\Rightarrow$ 'Ground is wet' is PROVEN.

Question 2 — Student Assignment Submission

Symbols:

- S = Student submits the assignment
- M = Student receives internal marks

1. Propositional Logic Expressions

- $S \rightarrow M$
- S (Rahul submitted the assignment)

2. CNF Conversion

- $S \rightarrow M \Rightarrow \neg S \vee M$
- S (already CNF)

Negate the Goal

Goal: $M \rightarrow$ Negated Goal: $\neg M$

Resolution Steps

#	Clause	Source
1	$\neg S \vee M$	Premise (CNF)
2	S	Fact
3	$\neg M$	Negated Goal

Step 1: Resolve (1) & (2): $\neg S \vee M + S \rightarrow M$

Step 2: Resolve M & $\neg M$ (3) $\rightarrow$ EMPTY CLAUSE ($\square$)

Conclusion: Contradiction derived $\Rightarrow$ 'Rahul receives internal marks' is PROVEN.

Question 3 — Library Membership

Symbols:

- L = Person is a library member
- B = Person can borrow books

1. Propositional Logic Representation

- $L \rightarrow B$
- L (Priya is a library member)

2. CNF Conversion

- $L \rightarrow B \Rightarrow \neg L \vee B$
- L (already CNF)

Negate the Goal

Goal: $B \rightarrow$ Negated Goal: $\neg B$

Resolution (until no new clauses can be generated)

#	Clause	Source
1	$\neg L \vee B$	Premise (CNF)
2	L	Fact
3	$\neg B$	Negated Goal

Step 1: Resolve (1) & (2): $\neg L \vee B + L \rightarrow B$

Step 2: Resolve B & $\neg B$ (3) $\rightarrow$ EMPTY CLAUSE ($\square$)

No further clauses can be generated — resolution terminates.

Final Conclusion: 'Priya can borrow books' is PROVEN.

Question 4 — Placement Eligibility

Symbols:

- C = Student clears the aptitude test
- E = Student is eligible for placement

1. Propositional Logic

- $C \rightarrow E$
- C (Arun cleared the aptitude test)

2. CNF Clauses

- $C \rightarrow E \Rightarrow \neg C \vee E$
- C (already CNF)

3. Negate the Goal

Goal: $E \rightarrow$ Negated Goal: $\neg E$

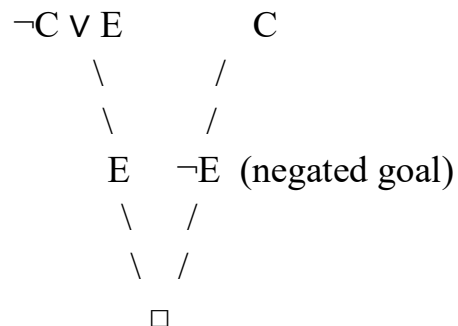
4. Resolution Steps

#	Clause	Source
1	$\neg C \vee E$	Premise (CNF)
2	C	Fact
3	$\neg E$	Negated Goal

Step 1: Resolve (1) & (2): $\neg C \vee E + C \rightarrow E$

Step 2: Resolve E & $\neg E$ (3) $\rightarrow$ EMPTY CLAUSE ($\square$)

5. Resolution Tree



Conclusion: Contradiction derived $\Rightarrow$ 'Arun is eligible for placement' is PROVEN.

Question 5 — Access Control System

Symbols:

- P = User enters the correct password
- A = User is authenticated
- G = User is granted access

1. Propositional Logic Representation

- $P \rightarrow A$
- $A \rightarrow G$
- P (the user entered the correct password)

2. CNF Conversion

- $P \rightarrow A \Rightarrow \neg P \vee A$
- $A \rightarrow G \Rightarrow \neg A \vee G$
- P (already CNF)

3. Negate the Goal

Goal: $G \rightarrow$ Negated Goal: $\neg G$

4. Resolution Steps (deriving the Empty Clause)

#	Clause	Source
1	$\neg P \vee A$	Premise (CNF)
2	$\neg A \vee G$	Premise (CNF)
3	P	Fact
4	$\neg G$	Negated Goal

Step 1: Resolve (1) & (3): $\neg P \vee A + P \rightarrow A$

Step 2: Resolve A(step1) & (2): $A + \neg A \vee G \rightarrow G$

Step 3: Resolve G & $\neg G$ (4) $\rightarrow$ EMPTY CLAUSE ($\square$)

5. Explanation of Each Step and Final Conclusion

- Step 1 combines 'password $\rightarrow$ authenticated' with the fact that the correct password was entered, deriving that the user IS authenticated (A).
- Step 2 combines 'authenticated $\rightarrow$ access granted' with the derived fact A, deriving that the user IS granted access (G).
- Step 3 resolves the derived G against the negated goal $\neg G$, producing a direct contradiction — the Empty Clause ($\square$).

Final Conclusion: Since assuming the goal is false leads to a contradiction, the original goal must be TRUE. 'The user is granted access' is PROVEN.